t3-w04 — Three Dimensional Shapes — Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Run `c0c3-deepseek-native` · conditions C0, C1, C2, C3 · the pack shown is draw 3 of each.

HER LESSON PLAN

Three Dimensional Shapes — Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Objectives

1. Identify the properties of cube and cuboids
2. Identify the properties of pyramids and cones

Step I — questions on previous knowledge

Teacher asks the students some questions based on their previous knowledge while the students answer and the teacher corrects their mistakes.

- **teacher** asks the students some questions based on their previous knowledge
- **student** answer
- **teacher** corrects their mistakes

Step II — properties of cubes and cuboids

Teacher writes the properties of cubes and cuboids on the board for the students to take notes in their notebooks.

- **teacher** writes the properties of cubes and cuboids on the board
- **student** take notes in their notebooks

Step III — properties of pyramids and cones

Teacher draws the pyramids and the cone on the board and writes their properties for the students to take down, as in Step II.

- **teacher** draws the pyramids and the cone on the board and writes their properties
- **student** take them down, as in Step II

Evaluation

Assembled from the plan's own two in-lesson activity sets, which between them cover both objectives. This is **not** the classwork the students sat — that was set from the textbook by reference (see `_question-set.md`). **Objective 1 — cubes and cuboids.** 1. How many vertices, has a cuboid? 2. How many edges has a cube? 3. Draw a cube and cuboid then label. **Objective 2 — pyramids and cones.** 4. Draw an hexagonal base pyramid and list the properties. 5. How many vertices has a tetrahedron.

1. How many vertices, has a cuboid?
2. How many edges has a cube?
3. Draw a cube and cuboid then label.
4. Draw an hexagonal base pyramid and list the properties.
5. How many vertices has a tetrahedron.

HER LESSON NOTE

Verbatim, as transcribed. **C2 and C3 receive this; C0 and C1 never do** — which is the whole of what separates them.

Step II — properties of cubes and cuboids

```
#### Three Dimensional shapes

These are shapes that have length, breadth and height. They are called solid shapes.

They may be either regular or irregular shapes. Regular shapes are cube of sugar, can drinks,
match box, etc. Irregular shapes are stone, trees etc.

#### Properties of cube

A cube drawn as a see-through solid: two equal squares, the nearer one offset down and to the
right of the farther one, with the four pairs of corresponding corners joined.

\begin{center}
\begin{tikzpicture}[scale=0.5]
  \draw (0,1.1) rectangle (4,5.1);
  \draw (1.1,0) rectangle (5.1,4);
  \draw (0,1.1)--(1.1,0);
  \draw (4,1.1)--(5.1,0);
  \draw (4,5.1)--(5.1,4);
  \draw (0,5.1)--(1.1,4);
\end{tikzpicture}
\end{center}

1. It is a solid state
2. It has 6 square faces
3. It has 12 edges
4. It has 8 vertices.

**Hint.** An edge is a line where two faces meet. It may be straight or curved. A face or
surface may be flat or curved. A vertex is a point corner where edges meet.

#### Properties of a Cuboid

A cuboid drawn the same way, wide and shallow, with three leader lines to the right naming the
top face, the nearer top-right corner and the nearer right-hand vertical edge.

\begin{center}
\begin{tikzpicture}[scale=0.5]
  \draw (0,1.05) rectangle (5,2.65);
  \draw (2.6,0) rectangle (7.6,1.6);
  \draw (0,1.05)--(2.6,0);
  \draw (5,1.05)--(7.6,0);
  \draw (5,2.65)--(7.6,1.6);
  \draw (0,2.65)--(2.6,1.6);
  \draw (4.3,2.4)--(8.5,2.4) node[right] {face};
  \draw (7.6,1.6)--(8.5,1.6) node[right] {vertex};
  \draw (7.6,0.8)--(8.5,0.8) node[right] {edge};
\end{tikzpicture}
\end{center}

1. It has 6 faces
```

```
2. It has 12 edges
3. It has 8 vertices.
```

Step III — properties of pyramids and cones

```
#### Pyramids

A pyramid is named based on its base shape.

A triangular base pyramid. The base triangle is drawn in perspective, its third vertex raised
into the middle of the figure, and the apex above is joined to all three base vertices.

\begin{center}
\begin{tikzpicture}[scale=0.55]
  \draw (2.05,0)--(5.45,0.25)--(3.75,2.4)--cycle;
  \draw (3.8,4.1)--(2.05,0);
  \draw (3.8,4.1)--(5.45,0.25);
  \draw (3.8,4.1)--(3.75,2.4);
  \node[align=center] at (3.6,-1.1) {Triangular base Pyramid\\(Tetrahedron)};
\end{tikzpicture}
\end{center}

A square base pyramid. The base is drawn as an upright rectangle and the apex above is joined
to all four of its corners, so two edges run outside the rectangle and two cross it.

\begin{center}
\begin{tikzpicture}[scale=0.55]
  \draw (0.7,1.9) rectangle (4,5);
  \draw (2.5,7.4)--(0.7,5);
  \draw (2.5,7.4)--(4,5);
  \draw (2.5,7.4)--(0.7,1.9);
  \draw (2.5,7.4)--(4,1.9);
  \node at (2.35,0.9) {Square base Pyramid};
\end{tikzpicture}
\end{center}

A pentagonal based pyramid. The base is drawn as a pentagon standing on one side with its
fifth vertex uppermost, and the apex above is joined to that vertex, to the two vertices
either side of it, and to the lower-right vertex — four edges from the apex, the last of them
crossing the face of the solid.

\begin{center}
\begin{tikzpicture}[scale=0.55]
  \draw (3.1,6.75)--(5.3,4.5)--(4.15,1.2)--(1.65,1.25)--(1.2,4.4)--cycle;
  \draw (3.3,9.65)--(1.2,4.4);
  \draw (3.3,9.65)--(3.1,6.75);
  \draw (3.3,9.65)--(5.3,4.5);
  \draw (3.3,9.65)--(4.15,1.2);
  \node[align=center] at (3,0.1) {Pentagonal based\\Pyramid};
\end{tikzpicture}
\end{center}

A rectangular based pyramid, drawn like the square one but on a base wider than it is tall.

\begin{center}
\begin{tikzpicture}[scale=0.55]
  \draw (1.15,2.3) rectangle (5,4.45);
  \draw (2.9,8)--(1.15,4.45);
  \draw (2.9,8)--(5,4.45);
```

```
  \draw (2.9,8)--(1.15,2.3);
  \draw (2.9,8)--(5,2.3);
  \node[align=center] at (3,1.2) {Rectangular based\\Pyramid};
\end{tikzpicture}
\end{center}

##### Properties

| Shape | Faces | edges | Vertices |
|---|---|---|---|
| Triangular based Pyramid | 4 | 6 | 4 |
| Square based Pyramid | 5 | 8 | 5 |
| Pentagonal base Pyramid | 6 | 10 | 6 |
| Rectangular based Pyramid | 5 | 8 | 5 |

#### Cone

Two drawings side by side. On the left an upright cone: an elliptical base with the two slant sides running up from the ends of the ellipse to the apex. On the right a cone lying on its side, apex at the left, its two slant sides reaching a narrow upright ellipse whose top and bottom then continue as two parallel lines to a second narrow ellipse further right.

\begin{center}
\begin{tikzpicture}[scale=0.5]
  \draw (0,0) ellipse (1.6 and 0.83);
  \draw (-1.6,0)--(-0.15,4.1)--(1.6,0);
  \begin{scope}[xshift=7cm]
    \draw (2.6,1)--(0,0)--(2.6,-1);
    \draw (2.6,0) ellipse (0.45 and 1);
    \draw (9.6,0) ellipse (0.5 and 1.05);
    \draw (2.6,1)--(9.6,1.05);
    \draw (2.6,-1)--(9.6,-1.05);
  \end{scope}
\end{tikzpicture}
\end{center}

1. It has two faces, i.e. one curved face and one flat face.
2. It has one edge.
3. It has one vertex
```

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three Dimensional Shapes

`subtopic`

Three dimensional figures: properties of cubes, cuboids, pyramids and cones

`term`

Third Term

`week`

4

`lesson_objectives`

• Identify the properties of cube and cuboids
• Identify the properties of pyramids and cones

Objective 1 — Identify the properties of cube and cuboids

`objective_text`

Identify the properties of cube and cuboids

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — Identify the properties of pyramids and cones

`objective_text`

Identify the properties of pyramids and cones

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Output — the pack it produced (draw 3)

Shapes That Fill Space: Cubes and Cuboids

EXPLANATION

Look around you and you will find shapes that are not flat. A dice, a box of matches, a shoe box — these are **three-dimensional** shapes: they have length, width and height, so they take up space. In this section you will meet two of the most familiar ones, the **cube** and the **cuboid**, and learn to describe them using three special words.

Every three-dimensional shape can be described by counting three things:

- **Faces** — the flat surfaces. If you ran your hand over the shape, each flat side you touch is one face.
- **Edges** — the straight lines where two faces meet.
- **Vertices** — the corners, where three or more edges meet. (One such corner is called a *vertex*.)

Let us apply these to a **cube**. A cube is a shape where every face is a square, and every face is exactly the same size. Think of a dice. A cube has:

- **6 faces**, all squares and all the same size,
- **12 edges**, all the same length,
- **8 vertices.**

Because every edge is the same length, a cube is a very "even" shape — all its sides are equal.

WORKED_EXAMPLE

Example: Describing a cube

Imagine a wooden dice. Count its faces, edges and vertices.

- **Faces:** Each side of the dice is a flat square. There is a top, a bottom, a front, a back, a left and a right — that makes **6 faces**.
- **Edges:** Along each square, there are edges where faces meet. Going around the top, around the bottom, and down each of the four corners in between, you count **12 edges**.
- **Vertices:** Each corner where three faces come together is a vertex. The dice has **8 corners**, so it has **8 vertices**.

So a cube has 6 faces, 12 edges and 8 vertices, and all 12 edges are equal in length.

EXPLANATION

Now meet the **cuboid** — think of a shoe box or a brick. A cuboid is like a cube that has been stretched: its faces are **rectangles**, and opposite faces are the same size, but the faces are not all the same as each other.

Count the same three things on a cuboid:

- **6 faces** — every face is a rectangle, and each face has a matching opposite face of the same size.
- **12 edges** — the edges come in sets of four equal edges: four for the length, four for the width and four for the height.
- **8 vertices.**

So a cube and a cuboid both have 6 faces, 12 edges and 8 vertices. The difference is their faces: a cube's faces are all equal squares, while a cuboid's faces are rectangles (opposite ones equal).

A quick way to remember: a **cube** is a special kind of cuboid where all three lengths are equal. When you hear "cube," picture the dice; when you hear "cuboid," picture the shoe box.

WORKED_EXAMPLE

Example: Comparing a cube and a cuboid

Look at a cube (dice) and a cuboid (shoe box) side by side.

Property	Cube (dice)	Cuboid (shoe box)
Faces	6 squares, all equal	6 rectangles, opposite ones equal
Edges	12, all equal in length	12, in three sets of four equal edges
Vertices	8	8

> Both have 6 faces, 12 edges and 8 vertices. The cube's faces are all the same squares; the cuboid's faces are rectangles that are not all the same.

1. A cuboid has 6 faces, 12 edges and how many vertices? Write the number.
answer: 8 vertices

2. What shape of faces does a cube have, and are they all the same size?
answer: A cube has 6 square faces, and they are all the same size.

3. State one difference between the faces of a cube and the faces of a cuboid.
answer: A cube's faces are all equal squares, while a cuboid's faces are rectangles (opposite faces equal).

4. A cuboid has 12 edges. How many of those edges are equal in length in each direction?
answer: 4 edges are equal in each direction (four for length, four for width, four for height).

Objective 2

Pyramids and Cones: Points, Faces and Edges

EXPLANATION

Now let's look at two more solid shapes: the pyramid and the cone.

What is a pyramid?
A pyramid is a solid with one flat base that is a polygon (a shape with straight sides, like a triangle or a square), and other faces that are triangles. All the triangular faces lean inward and meet at a single point at the top, called the apex (or vertex).

A pyramid is named after the shape of its base:

A triangular pyramid has a triangular base.
A square pyramid has a square base.
A rectangular pyramid has a rectangular base.
A pentagonal pyramid has a pentagonal base.

The number of faces, edges and vertices of a pyramid depends on the number of sides of its base. Suppose the base has n sides:

Faces: n + 1 (the base, plus n triangular faces).
Vertices (corners): n + 1 (the n corners of the base, plus the apex).
Edges: 2n (the n edges around the base, plus the n edges that run up to the apex).

EXPLANATION

What is a cone?
A cone is a solid with one flat circular base and one curved surface that tapers smoothly up to a single point at the top, also called the apex (or vertex).

A cone always has exactly:

1 face that is flat (the circular base).
1 curved surface that is not flat.
1 vertex (the apex at the top).
1 edge (the circular rim where the base meets the curved surface).

The sharp point of a cone is its apex, and the round rim at the bottom is its single edge. Unlike a pyramid, whose base is a polygon with straight sides, a cone's base is a circle, so its base has just one continuous curved edge.

WORKED_EXAMPLE

Worked Example 1: Counting the parts of a square pyramid

A square pyramid has a square base. How many faces, vertices and edges does it have?

Step 1 — Find the number of faces.
The base has n = 4 sides. Faces = n + 1 = 4 + 1 = 5.
The square pyramid has 5 faces: the square base and 4 triangular faces.

Step 2 — Find the number of vertices.
Vertices = n + 1 = 4 + 1 = 5.
It has 5 vertices: the 4 corners of the square base and the apex at the top.

Step 3 — Find the number of edges.
Edges = 2n = 2 × 4 = 8.
It has 8 edges: the 4 edges around the square base and the 4 edges sloping up to the apex.

So a square pyramid has 5 faces, 5 vertices and 8 edges.

> WORKED_EXAMPLE
> Worked Example 2: Counting the parts of a cone
>
> A cone has a circular base. How many faces, vertices and edges does it have?
>
> Step 1 — Faces.
> A cone has 1 flat circular base and 1 curved surface, so it has 1 flat face (plus 1 curved surface that is not a face).
>
> Step 2 — Vertices.
> It has 1 vertex, the apex at the top.
>
> Step 3 — Edges.
> It has 1 edge, the circular rim where the base meets the curved surface.
>
> So a cone has 1 flat face, 1 vertex and 1 edge (plus 1 curved surface).

1. A triangular pyramid has a triangular base. How many faces, vertices and edges does it have?
answer: 4 faces, 4 vertices and 6 edges.

2. A pentagonal pyramid has a base with 5 sides. How many faces and how many vertices does it have?
answer: 6 faces and 6 vertices.

3. A cone has a circular base. How many flat faces, vertices and edges does it have?
answer: 1 flat face, 1 vertex and 1 edge (plus 1 curved surface).

4. A square pyramid and a cone each have exactly one apex. In what other way is a square pyramid different from a cone?
answer: A square pyramid has a square (polygon) base with 5 faces, 5 vertices and 8 edges, all its non-base faces are triangles; a cone has a circular base with 1 flat face, 1 vertex and 1 edge, and its side is a single curved surface.

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three Dimensional Shapes

`subtopic`

Three dimensional figures: properties of cubes, cuboids, pyramids and cones

`term`

Third Term

`week`

4

`lesson_objectives`

• Identify the properties of cube and cuboids
• Identify the properties of pyramids and cones

`entry_behaviour`

• Students have been taught the differences between cube and cuboid.

`instructional_materials`

• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.

Objective 1 — Identify the properties of cube and cuboids

`objective_text`

Identify the properties of cube and cuboids

`current_step`

1

`instructional_events`

• present_content

`assessment_points`

• How many vertices, has a cuboid?
• How many edges has a cube?
• Draw a cube and cuboid then label.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — Identify the properties of pyramids and cones

`objective_text`

Identify the properties of pyramids and cones

`current_step`

2

`instructional_events`

• present_content

`assessment_points`

• Draw an hexagonal base pyramid and list the properties.
• How many vertices has a tetrahedron.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

The Properties of Cubes and Cuboids

EXPLANATION

A **cube** and a **cuboid** are both solid (three-dimensional) shapes, and they are closely related. The best way to tell them apart is to look at their **faces** (the flat surfaces), **edges** (the lines where two faces meet) and **vertices** (the corners, where three or more edges meet).

Think of a cube like a **die**. Every face of a cube is a **square**, and all its edges are **equal** in length. So a cube has:

- **6 faces**, and every one of them is a square;
- **12 edges**, all equal in length;
- **8 vertices**.

A **cuboid** is like a **box** — think of a shoe box or a brick. Its faces are **rectangles**, not all squares. It has:

- **6 faces**, and they are rectangles;
- **12 edges**;
- **8 vertices**.

Notice that both shapes have the **same number** of faces, edges and vertices — 6, 12 and 8. What makes them different is their **faces**. A cube's six faces are all squares; a cuboid's six faces are rectangles. A cube is really a special kind of cuboid whose edges all happen to be equal.

WORKED_EXAMPLE

Let's work through an example together so you can see how to answer questions about these properties.

Example: A cuboid has how many vertices?

Remember that a cuboid is a box shape. Each of its corners is a vertex. Count them: there are four vertices on the top and four on the bottom, which makes **8 vertices** altogether.

Example: How many edges does a cube have?

A cube is a die shape. Count the edges along the top (4), the bottom (4) and the four edges joining the top to the bottom (4). That gives 4 + 4 + 4 = **12 edges**.

Now let's look at how to decide whether a shape is a cube or a cuboid. Look at all six faces. If every face is a **square**, it is a **cube**. If the faces are **rectangles**, it is a **cuboid**.

1. How many faces, edges and vertices does a cube have? Write all three numbers.
answer: A cube has 6 faces, 12 edges and 8 vertices.

2. How many vertices does a cuboid have?
answer: A cuboid has 8 vertices.

3. How many edges does a cube have?
answer: A cube has 12 edges.

4. A solid shape has six faces, and every one of its faces is a square. Is it a cube or a cuboid?
answer: It is a cube, because all of a cube's faces are squares.

5. How are the faces of a cuboid different from the faces of a cube?
answer: The faces of a cuboid are rectangles, while the faces of a cube are all squares.

Objective 2

Pyramids and Cones: Naming Every Part

EXPLANATION

A pyramid is a solid whose base is a flat polygon (a shape with straight sides, such as a triangle, square or pentagon) and whose other faces are triangles that all meet at one point at the top. That top point is called the apex. A cone is a solid with one flat circular base and one curved surface that slopes up to a single point, also called the apex.

Let's name every part of a pyramid so you can count its properties. Take a pyramid whose base is a pentagon (a five-sided shape).

• The base is one face — the pentagon at the bottom.
• Each side of the pentagon carries one triangular face, so five triangular faces slope up to the apex.
• Faces: base (1) + five triangles (5) = 6 faces.
• Vertices (corners): the five corners of the pentagon base plus the apex = 6 vertices.
• Edges: the five edges of the base plus the five edges running from the base up to the apex = 10 edges.

So the pattern for any pyramid is easy: if the base has n sides, then the pyramid has n + 1 faces, n + 1 vertices, and 2n edges. The "+1" every time is the apex and the triangular faces that meet there.

A special pyramid is the tetrahedron — a pyramid whose base is a triangle. With a three-sided base, it has 3 + 1 = 4 faces, 3 + 1 = 4 vertices, and 2 × 3 = 6 edges.

Now for the cone. A cone is different from a pyramid because its base is a circle, not a polygon.

• It has one flat circular base face.
• It has one curved surface that rises to the apex.
• Faces: 2 (the base and the curved surface).
• Vertices: 1 — only the apex, because the circular base has no corners.
• Edges: 1 — the circular rim where the base meets the curved surface.

WORKED_EXAMPLE

Worked example — a pyramid on a hexagonal base.

A hexagonal base has six sides, so n = 6.

Faces: n + 1 = 6 + 1 = 7. The pyramid has six triangular side faces and the hexagonal base.
Vertices: n + 1 = 6 + 1 = 7. Six corners of the base plus the apex.
Edges: 2n = 2 × 6 = 12. Six base edges plus six sloping edges to the apex.

So a hexagonal-base pyramid has 7 faces, 7 vertices and 12 edges.

Worked example — the tetrahedron.

Its base is a triangle, so n = 3.

Faces: 3 + 1 = 4 (three triangular sides plus the triangular base).
Vertices: 3 + 1 = 4.
Edges: 2 × 3 = 6.

So a tetrahedron has 4 faces, 4 vertices and 6 edges. Notice the two counts match — that is why it is sometimes called a "four-faced" solid.

1. A pyramid has a base that is a heptagon (a seven-sided polygon). How many faces, vertices and edges does it have?
answer: 8 faces, 8 vertices, 14 edges

2. How many vertices does a tetrahedron have?
answer: 4 vertices

3. A cone is placed on a table. List the number of faces, vertices and edges of the cone.
answer: 2 faces (one circular base and one curved surface), 1 vertex (the apex), 1 edge (the circular rim)

C2 `C1 plus the lesson note as source evidence.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three Dimensional Shapes

`subtopic`

Three dimensional figures: properties of cubes, cuboids, pyramids and cones

`term`

Third Term

`week`

4

`lesson_objectives`

• Identify the properties of cube and cuboids
• Identify the properties of pyramids and cones

`entry_behaviour`

• Students have been taught the differences between cube and cuboid.

`instructional_materials`

• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.

Objective 1 — Identify the properties of cube and cuboids

`objective_text`

Identify the properties of cube and cuboids

`current_step`

1

`instructional_events`

• present_content

`assessment_points`

• How many vertices, has a cuboid?
• How many edges has a cube?
• Draw a cube and cuboid then label.

`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Three Dimensional shapes\n\nThese are shapes that have length, breadth and
height. They are called solid shapes.\n\nThey may be either regular or irregular shapes. Regular
shapes are cube of sugar, can drinks,\nmatch box, etc. Irregular shapes are stone, trees etc.
\n\n#### Properties of cube\n\nA cube drawn as a see-through solid: two equal squares, the nearer
one offset down and to the\nright of the farther one, with the four pairs of corresponding
corners joined.\n\n[drawing attached to this section here, unlabelled]\n\n1. It is a solid
state\n2. It has 6 square faces\n3. It has 12 edges\n4. It has 8 vertices.\n\n**Hint.** An edge
is a line where two faces meet. It may be straight or curved. A face or\nsurface may be flat or
curved. A vertex is a point corner where edges meet.\n\n#### Properties of a Cuboid\n\nA cuboid
drawn the same way, wide and shallow, with three leader lines to the right naming the\ntop face,
the nearer top-right corner and the nearer right-hand vertical edge.\n\n[drawing attached to this
section here, unlabelled, marked face, vertex, edge]\n\n1. It has 6 faces\n2. It has 12 edges\n3.
It has 8 vertices."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — Identify the properties of pyramids and cones

`objective_text`

Identify the properties of pyramids and cones

`current_step`

2

`instructional_events`

• present_content

`assessment_points`

• Draw an hexagonal base pyramid and list the properties.
• How many vertices has a tetrahedron.

`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Pyramids\n\nA pyramid is named based on its base shape.\n\nA triangular base
pyramid. The base triangle is drawn in perspective, its third vertex raised\ninto the middle of
the figure, and the apex above is joined to all three base vertices.\n\n[drawing attached to this
section here, unlabelled, marked Triangular base Pyramid (Tetrahedron)]\n\nA square base pyramid.
The base is drawn as an upright rectangle and the apex above is joined\nto all four of its
corners, so two edges run outside the rectangle and two cross it.\n\n[drawing attached to this
section here, unlabelled, marked Square base Pyramid]\n\nA pentagonal based pyramid. The base is
drawn as a pentagon standing on one side with its\nfifth vertex uppermost, and the apex above is
joined to that vertex, to the two vertices\neither side of it, and to the lower-right vertex —
four edges from the apex, the last of them\ncrossing the face of the solid.\n\n[drawing attached
to this section here, unlabelled, marked Pentagonal based Pyramid]\n\nA rectangular based
pyramid, drawn like the square one but on a base wider than it is tall.\n\n[drawing attached to
this section here, unlabelled, marked Rectangular based Pyramid]\n\n##### Properties\n\n| Shape |
Faces | edges | Vertices |\n|---|---|---|---|\n| Triangular based Pyramid | 4 | 6 | 4 |\n| Square
based Pyramid | 5 | 8 | 5 |\n| Pentagonal base Pyramid | 6 | 10 | 6 |\n| Rectangular based
Pyramid | 5 | 8 | 5 |\n\n#### Cone\n\nTwo drawings side by side. On the left an upright cone: an
elliptical base with the two slant\nsides running up from the ends of the ellipse to the apex. On
the right a cone lying on its\nside, apex at the left, its two slant sides reaching a narrow
upright ellipse whose top and\nbottom then continue as two parallel lines to a second narrow
ellipse further right.\n\n[drawing attached to this section here, unlabelled]\n\n1. It has two
faces, i.e. one curved face and one flat face.\n2. It has one edge.\n3. It has one vertex"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Cubes and Cuboids: Faces, Edges and Vertices

EXPLANATION

Shapes that have length, breadth **and** height are called **three-dimensional** (or **solid**) shapes. You can think of them as shapes that take up space.

Every solid shape is built from three parts, and it helps to know their names before you count:

- A **face** (or surface) is a flat or curved surface of the solid.
- An **edge** is a line where two faces meet. It may be straight or curved.
- A **vertex** is a point (corner) where edges meet. (More than one are called **vertices**.)

Look at fig-01, which shows a cube drawn as a see-through solid. A **cube** has these properties:

1. It is a solid shape.
2. It has **6 square faces** — all six faces are squares, and they are all the same size.
3. It has **12 edges**.
4. It has **8 vertices**.

Now look at fig-02, which shows a cuboid. Its leader lines point to one face, one vertex and one edge so you can see each part. A **cuboid** has these properties:

1. It has **6 faces** (its faces are rectangles, so it looks like a stretched or flattened cube).
2. It has **12 edges**.
3. It has **8 vertices**.

So a cube and a cuboid both have 6 faces, 12 edges and 8 vertices — the difference is that a cube's faces are all equal squares, while a cuboid's faces are rectangles.

VISUAL

```
\begin{tikzpicture}[scale=0.5]
  \draw (0,1.05) rectangle (5,2.65);
  \draw (2.6,0) rectangle (7.6,1.6);
  \draw (0,1.05)--(2.6,0);
  \draw (5,1.05)--(7.6,0);
  \draw (5,2.65)--(7.6,1.6);
  \draw (0,2.65)--(2.6,1.6);
  \draw (4.3,2.4)--(8.5,2.4) node[right] {face};
  \draw (7.6,1.6)--(8.5,1.6) node[right] {vertex};
  \draw (7.6,0.8)--(8.5,0.8) node[right] {edge};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.5]
  \draw (0,1.1) rectangle (4,5.1);
  \draw (1.1,0) rectangle (5.1,4);
  \draw (0,1.1)--(1.1,0);
  \draw (4,1.1)--(5.1,0);
  \draw (4,5.1)--(5.1,4);
  \draw (0,5.1)--(1.1,4);
\end{tikzpicture}
```

WORKED_EXAMPLE

Worked example — counting the parts of a cuboid

Use fig-02 to check the properties of the cuboid.

- **Faces.** Count the flat surfaces that make up the cuboid: there are 6 — a top face and a bottom face, plus four faces around the sides. The leader line in fig-02 names the top face.
- **Edges.** An edge is a line where two faces meet. Count the straight lines that form the frame of the cuboid: 4 on the top face, 4

> on the bottom face, and 4 vertical lines joining them — that is **12 edges** in total. The leader line in fig-02 names one of the vertical edges.
> - **Vertices.** A vertex is a corner where edges meet. The cuboid has a corner on top and a corner at the bottom of each of its four corners — that is **8 vertices**. The leader line in fig-02 names one vertex.
>
> So the cuboid has 6 faces, 12 edges and 8 vertices.

1. How many vertices does a cuboid have?
answer: 8 vertices

2. How many edges does a cube have?
answer: 12 edges

3. How many faces does a cube have, and what shape is each face?
answer: 6 faces, and each face is a square.

4. A cube and a cuboid both have the same number of faces, edges and vertices. What is the difference between them?
answer: A cube's six faces are all equal squares, while a cuboid's faces are rectangles.

Pyramids and Cones: Their Properties

EXPLANATION

A pyramid is a three-dimensional shape with a flat base and sloping triangular faces that meet at a single point at the top, called the apex. A pyramid is named after the shape of its base — so a base that is a triangle gives a triangular-based pyramid, a square base gives a square-based pyramid, and so on.

Every pyramid has three things you can count: its faces (the flat surfaces), its edges (where two faces meet) and its vertices (the corners, where edges meet). You can read these straight off the drawing. Look at fig-04, the square-based pyramid: the base has four corners and the apex sits above it, and the apex is joined to all four corners of the base.

VISUAL

```
\begin{tikzpicture}[scale=0.55]
  \draw (0.7,1.9) rectangle (4,5);
  \draw (2.5,7.4)--(0.7,5);
  \draw (2.5,7.4)--(4,5);
  \draw (2.5,7.4)--(0.7,1.9);
  \draw (2.5,7.4)--(4,1.9);
  \node at (2.35,0.9) {Square base Pyramid};
\end{tikzpicture}
```

EXPLANATION

Here are the properties of the pyramids you need to know. Faces are the flat surfaces, edges are the lines where faces meet, and vertices are the corners.

Pyramid	Faces	Edges	Vertices
Triangular-based pyramid	4	6	4
Square-based pyramid	5	8	5
Pentagonal-based pyramid	6	10	6
Rectangular-based pyramid	5	8	5

Notice the pattern: a pyramid whose base has n sides has n + 1 faces, 2n edges and n + 1 vertices. The base provides one face, and each side of the base gives one sloping triangular face, which together with the base gives n + 1 faces in all.

A triangular-based pyramid has only four faces, so it is also called a tetrahedron ("tetra" means four).

EXPLANATION

A cone is different from a pyramid. Instead of a flat-sided base, a cone has a circular base and a single curved surface that narrows to a point at the top (the apex). Look at fig-07: the upright cone has a flat elliptical base and two slant sides meeting at the apex, and the lying cone shows the same parts from the side.

The cone has just three properties to remember:

1. It has two faces — one curved face and one flat face.
2. It has one edge (the rim where the curved face meets the flat circular base).
3. It has one vertex (the point at the top).

VISUAL

```
\begin{tikzpicture}[scale=0.5]
  \draw (0,0) ellipse (1.6 and 0.83);
  \draw (-1.6,0)--(-0.15,4.1)--(1.6,0);
  \begin{scope}[xshift=7cm]
    \draw (2.6,1)--(0,0)--(2.6,-1);
    \draw (2.6,0) ellipse (0.45 and 1);
    \draw (9.6,0) ellipse (0.5 and 1.05);
    \draw (2.6,1)--(9.6,1.05);
    \draw (2.6,-1)--(9.6,-1.05);
  \end{scope}
\end{tikzpicture}
```

WORKED_EXAMPLE
Worked example — the square-based pyramid

Take the square-based pyramid in fig-04 and count its properties.

Its base is a square, which is one flat face. Each of the four sides of that square rises to the apex as a triangular face, giving four sloping triangular faces. Altogether that is 1 base face + 4 sloping faces = 5 faces.

For the edges: the base has 4 edges around its square, and from each of the 4 corners of the base an edge runs up to the apex, giving 4 more. That is 4 + 4 = 8 edges.

For the vertices: there are the 4 corners of the base, plus the apex at the top = 5 vertices.

So a square-based pyramid has 5 faces, 8 edges and 5 vertices — exactly matching the row in the table.

VISUAL
```
\begin{tikzpicture}[scale=0.55]
  \draw (2.05,0)--(5.45,0.25)--(3.75,2.4)--cycle;
  \draw (3.8,4.1)--(2.05,0);
  \draw (3.8,4.1)--(5.45,0.25);
  \draw (3.8,4.1)--(3.75,2.4);
  \node[align=center] at (3.6,-1.1) {Triangular base Pyramid\\(Tetrahedron)};
\end{tikzpicture}
```

VISUAL
```
\begin{tikzpicture}[scale=0.55]
  \draw (3.1,6.75)--(5.3,4.5)--(4.15,1.2)--(1.65,1.25)--(1.2,4.4)--cycle;
  \draw (3.3,9.65)--(1.2,4.4);
  \draw (3.3,9.65)--(3.1,6.75);
  \draw (3.3,9.65)--(5.3,4.5);
  \draw (3.3,9.65)--(4.15,1.2);
  \node[align=center] at (3,0.1) {Pentagonal based\\Pyramid};
\end{tikzpicture}
```

VISUAL
```
\begin{tikzpicture}[scale=0.55]
  \draw (1.15,2.3) rectangle (5,4.45);
  \draw (2.9,8)--(1.15,4.45);
  \draw (2.9,8)--(5,4.45);
  \draw (2.9,8)--(1.15,2.3);
  \draw (2.9,8)--(5,2.3);
  \node[align=center] at (3,1.2) {Rectangular based\\Pyramid};
\end{tikzpicture}
```

1. A pentagonal-based pyramid has a base with five sides. Using the pattern, how many faces, edges and vertices does it have?
answer: It has 6 faces, 10 edges and 6 vertices.

2. How many vertices does a tetrahedron have?
answer: A tetrahedron has 4 vertices.

3. List all the properties of a cone — its faces, its edge and its vertex.
answer: A cone has 2 faces (one curved and one flat), 1 edge and 1 vertex.

4. A pyramid has 5 faces in total. What shape is its base, and how many edges does it have?
answer: Its base is a square (or a rectangle), and it has 8 edges.

C3 `C2 plus a validation and repair pass over the draft.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three Dimensional Shapes

`subtopic`

Three dimensional figures: properties of cubes, cuboids, pyramids and cones

`term`

Third Term

`week`

4

`lesson_objectives`

• Identify the properties of cube and cuboids
• Identify the properties of pyramids and cones

`entry_behaviour`

• Students have been taught the differences between cube and cuboid.

`instructional_materials`

• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.

Objective 1 — Identify the properties of cube and cuboids

objective_text

Identify the properties of cube and cuboids

current_step

1

instructional_events

• present_content

assessment_points

• How many vertices, has a cuboid?
• How many edges has a cube?
• Draw a cube and cuboid then label.

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Three Dimensional shapes\n\nThese are shapes that have length, breadth and
height. They are called solid shapes.\n\nThey may be either regular or irregular shapes. Regular
shapes are cube of sugar, can drinks,\nmatch box, etc. Irregular shapes are stone, trees etc.
\n\n#### Properties of cube\n\nA cube drawn as a see-through solid: two equal squares, the nearer
one offset down and to the\nright of the farther one, with the four pairs of corresponding
corners joined.\n\n[drawing attached to this section here, unlabelled]\n\n1. It is a solid
state\n2. It has 6 square faces\n3. It has 12 edges\n4. It has 8 vertices.\n\n**Hint.** An edge
is a line where two faces meet. It may be straight or curved. A face or\nsurface may be flat or
curved. A vertex is a point corner where edges meet.\n\n#### Properties of a Cuboid\n\nA cuboid
drawn the same way, wide and shallow, with three leader lines to the right naming the\ntop face,
the nearer top-right corner and the nearer right-hand vertical edge.\n\n[drawing attached to this
section here, unlabelled, marked face, vertex, edge]\n\n1. It has 6 faces\n2. It has 12 edges\n3.
It has 8 vertices."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — Identify the properties of pyramids and cones

`objective_text`

Identify the properties of pyramids and cones

`current_step`

2

`instructional_events`

• present_content

`assessment_points`

• Draw an hexagonal base pyramid and list the properties.
• How many vertices has a tetrahedron.

`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Pyramids\n\nA pyramid is named based on its base shape.\n\nA triangular base
pyramid. The base triangle is drawn in perspective, its third vertex raised\ninto the middle of
the figure, and the apex above is joined to all three base vertices.\n\n[drawing attached to this
section here, unlabelled, marked Triangular base Pyramid (Tetrahedron)]\n\nA square base pyramid.
The base is drawn as an upright rectangle and the apex above is joined\nto all four of its
corners, so two edges run outside the rectangle and two cross it.\n\n[drawing attached to this
section here, unlabelled, marked Square base Pyramid]\n\nA pentagonal based pyramid. The base is
drawn as a pentagon standing on one side with its\nfifth vertex uppermost, and the apex above is
joined to that vertex, to the two vertices\neither side of it, and to the lower-right vertex —
four edges from the apex, the last of them\ncrossing the face of the solid.\n\n[drawing attached
to this section here, unlabelled, marked Pentagonal based Pyramid]\n\nA rectangular based
pyramid, drawn like the square one but on a base wider than it is tall.\n\n[drawing attached to
this section here, unlabelled, marked Rectangular based Pyramid]\n\n##### Properties\n\n| Shape |
Faces | edges | Vertices |\n|---|---|---|---|\n| Triangular based Pyramid | 4 | 6 | 4 |\n| Square
based Pyramid | 5 | 8 | 5 |\n| Pentagonal base Pyramid | 6 | 10 | 6 |\n| Rectangular based
Pyramid | 5 | 8 | 5 |\n\n#### Cone\n\nTwo drawings side by side. On the left an upright cone: an
elliptical base with the two slant\nsides running up from the ends of the ellipse to the apex. On
the right a cone lying on its\nside, apex at the left, its two slant sides reaching a narrow
upright ellipse whose top and\nbottom then continue as two parallel lines to a second narrow
ellipse further right.\n\n[drawing attached to this section here, unlabelled]\n\n1. It has two
faces, i.e. one curved face and one flat face.\n2. It has one edge.\n3. It has one vertex"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Cubes and Cuboids: What Makes a Solid Shape

EXPLANATION
Three-dimensional shapes are shapes that have length, breadth and height, so they are called solid shapes. They can be regular or irregular. Regular shapes include a cube of sugar, a can of drink and a match box; irregular shapes include a stone or a tree.

To describe a solid shape we count three kinds of parts. A face (or surface) is a flat or curved side of the shape. An edge is a line where two faces meet, and it may be straight or curved. A vertex is a point corner where edges meet.

EXPLANATION
A cube is a solid shape in which every face is an equal square. Look at the cube in fig-01: it is drawn see-through so you can see the square faces and the edges joining them.

The properties of a cube are:

It is a solid shape.
It has 6 square faces.
It has 12 edges.
It has 8 vertices.

Because every face of a cube is an equal square, all twelve edges are the same length.

VISUAL
```
\begin{tikzpicture}[scale=0.5]
  \draw (0,1.1) rectangle (4,5.1);
  \draw (1.1,0) rectangle (5.1,4);
  \draw (0,1.1)--(1.1,0);
  \draw (4,1.1)--(5.1,0);
  \draw (4,5.1)--(5.1,4);
  \draw (0,5.1)--(1.1,4);
\end{tikzpicture}
```

EXPLANATION
A cuboid is a solid shape like a box: it is wide and shallow rather than perfectly square. Fig-02 shows a cuboid with its top face, a vertex (a point corner) and an edge (a line where two faces meet) named.

The properties of a cuboid are:

It has 6 faces.
It has 12 edges.
It has 8 vertices.

The faces of a cuboid are rectangles, so its edges are not all the same length.

VISUAL
```
\begin{tikzpicture}[scale=0.5]
  \draw (0,1.05) rectangle (5,2.65);
  \draw (2.6,0) rectangle (7.6,1.6);
  \draw (0,1.05)--(2.6,0);
  \draw (5,1.05)--(7.6,0);
  \draw (5,2.65)--(7.6,1.6);
  \draw (0,2.65)--(2.6,1.6);
  \draw (4.3,2.4)--(8.5,2.4) node[right] {face};
  \draw (7.6,1.6)--(8.5,1.6) node[right] {vertex};
  \draw (7.6,0.8)--(8.5,0.8) node[right] {edge};
\end{tikzpicture}
```

1. How many vertices has a cuboid?
answer: 8

2. How many edges has a cube?
answer: 12

3. Copy the cube in fig-01 and the cuboid in fig-02, then label on each solid: one face, one edge and one vertex.
answer: Your drawing should mark one face (a flat square or rectangle side), one edge (a line where two faces meet) and one vertex (a point corner where edges meet) on each solid.

Pyramids and Cones: Their Properties

EXPLANATION

A pyramid is named after the shape of its base. So a pyramid whose base is a triangle is called a triangular based pyramid, one whose base is a square is a square based pyramid, and so on. Every pyramid has one flat base and triangular faces that slope up to a single point called the apex.

Each kind of pyramid has its own count of faces, edges and vertices, and you can find the count from the shape of the base. The triangular based pyramid is sometimes called a tetrahedron, because it has four faces.

EXPLANATION

Here are the properties of the common pyramids. Study them, then use them in the practice below.

Shape	Faces	Edges	Vertices
Triangular based pyramid (tetrahedron)	4	6	4
Square based pyramid	5	8	5
Pentagonal based pyramid	6	10	6
Rectangular based pyramid	5	8	5

Notice the pattern: for a pyramid with a base of n sides, it has n + 1 faces, 2n edges and n + 1 vertices. For example, a pentagonal base has 5 sides, so it has 5 + 1 = 6 faces, 2 × 5 = 10 edges and 5 + 1 = 6 vertices, just as the table shows.

EXPLANATION

A cone is a different kind of three dimensional shape. It has a flat circular base and a single curved surface that narrows to a point at the top, called the apex. Its properties are:

1. It has two faces — one curved face and one flat face.
2. It has one edge.
3. It has one vertex.

So a cone has far fewer parts than a pyramid: just two faces, one edge and one vertex.

VISUAL

```
\begin{tikzpicture}[scale=0.55]
  \draw (2.05,0)--(5.45,0.25)--(3.75,2.4)--cycle;
  \draw (3.8,4.1)--(2.05,0);
  \draw (3.8,4.1)--(5.45,0.25);
  \draw (3.8,4.1)--(3.75,2.4);
  \node[align=center] at (3.6,-1.1) {Triangular base Pyramid\\(Tetrahedron)};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55]
  \draw (0.7,1.9) rectangle (4,5);
  \draw (2.5,7.4)--(0.7,5);
  \draw (2.5,7.4)--(4,5);
  \draw (2.5,7.4)--(0.7,1.9);
  \draw (2.5,7.4)--(4,1.9);
  \node at (2.35,0.9) {Square base Pyramid};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55]
  \draw (3.1,6.75)--(5.3,4.5)--(4.15,1.2)--(1.65,1.25)--(1.2,4.4)--cycle;
  \draw (3.3,9.65)--(1.2,4.4);
  \draw (3.3,9.65)--(3.1,6.75);
  \draw (3.3,9.65)--(5.3,4.5);
  \draw (3.3,9.65)--(4.15,1.2);
  \node[align=center] at (3,0.1) {Pentagonal based\\Pyramid};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55]
  \draw (1.15,2.3) rectangle (5,4.45);
  \draw (2.9,8)--(1.15,4.45);
  \draw (2.9,8)--(5,4.45);
  \draw (2.9,8)--(1.15,2.3);
  \draw (2.9,8)--(5,2.3);
  \node[align=center] at (3,1.2) {Rectangular based\\Pyramid};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.5]
  \draw (0,0) ellipse (1.6 and 0.83);
  \draw (-1.6,0)--(-0.15,4.1)--(1.6,0);
  \begin{scope}[xshift=7cm]
    \draw (2.6,1)--(0,0)--(2.6,-1);
    \draw (2.6,0) ellipse (0.45 and 1);
    \draw (9.6,0) ellipse (0.5 and 1.05);
    \draw (2.6,1)--(9.6,1.05);
    \draw (2.6,-1)--(9.6,-1.05);
  \end{scope}
\end{tikzpicture}
```

1. A pyramid has a square base. How many faces, edges and vertices does it have?
answer: 5 faces, 8 edges, 5 vertices.

2. How many vertices has a tetrahedron?
answer: 4 vertices.

3. State the number of faces, edges and vertices of a cone.
answer: 2 faces (one curved, one flat), 1 edge, 1 vertex.

4. A pyramid has a hexagonal base. Using the pattern for a base of n sides, list its number of faces, edges and vertices.
answer: A hexagonal base has 6 sides, so: faces = 6 + 1 = 7, edges = 2 × 6 = 12, vertices = 6 + 1 = 7. So a hexagonal based pyramid has 7 faces, 12 edges and 7 vertices.

5. List the faces, edges and vertices of a triangular based pyramid.
answer: A triangular based pyramid (tetrahedron) has 4 faces, 6 edges and 4 vertices.

Result — per rubric dimension

Scores are 1–5. Every figure is the mean of 3 draws rated by 2 judge pass(es) — 6 judgments per cell.

Rubric	C0	C1	C2	C3	C1 – C0	C2 – C1	C3 – C2
Objective and curriculum alignment	5.00	5.00	5.00	5.00	+0.00	+0.00	+0.00
Taught-lesson alignment	3.67	4.00	4.00	4.00	**+0.33**	+0.00	+0.00
Scope and conceptual correctness	4.67	5.00	4.83	4.83	**+0.33**	**-0.17**	+0.00
Student-facing clarity and usefulness	5.00	5.00	5.00	5.00	+0.00	+0.00	+0.00

Why the judge scored it that way

Every rationale the judge wrote for this lesson, deduplicated across draws and passes. These are its words, not a summary.

Objective and curriculum alignment — C0 5.00, C1 5.00, C2 5.00, C3 5.00

- `C0` Both lesson objectives are covered completely and in the correct order. Every property named in the objectives is present in the revision pack, making alignment essentially perfect.
- `C0` The pack covers both stated lesson objectives completely and in sufficient depth for JSS1, and the content maps directly onto the curriculum target of 3D shape properties.
- `C0` Both lesson objectives are fully addressed: the pack identifies and explains the properties of cubes and cuboids, then pyramids and cones, in direct correspondence with the stated curriculum targets for JSS1 Third Term week 4.
- `C0` The revision pack covers both lesson objectives completely and accurately, matching the JSS1 curriculum target for this topic. Nothing relevant is omitted and nothing extraneous is introduced.
- `C0` The pack addresses both stated objectives thoroughly and stays firmly within the JSS1 third-term curriculum target on three-dimensional shapes. All key properties (faces, edges, vertices) are covered for all four required shapes.
- `C0` Both stated lesson objectives are comprehensively covered in the revision pack. The pack addresses all key properties (faces, edges, vertices) for cubes, cuboids, pyramids and cones, matching the JSS1 Third Term curriculum target for three-dimensional shapes.
- `C1` The pack covers both stated lesson objectives completely and at the expected JSS1 depth, addressing every shape named in the lesson plan (cube, cuboid, pyramid, cone) and their key properties. Alignment with the curriculum target is thorough.
- `C1` Both lesson objectives are covered completely and in the correct order, with content depth appropriate to JSS1. The pack aligns tightly with the stated curriculum target.
- `C1` Both lesson objectives are met comprehensively and in the correct JSS1 scope. The pack covers every shape named in the lesson plan — cube, cuboid, pyramid (including hexagonal-base and tetrahedron), and cone — with accurate properties stated for each.
- `C1` The pack directly addresses both stated lesson objectives and maps onto the teacher's evaluation questions, demonstrating strong alignment with the intended curriculum target for JSS1 Term 3 Week 4.
- `C1` Both lesson objectives are completely addressed with appropriate depth for JSS1, and every teacher evaluation question finds a direct counterpart in the revision pack's practice items and solutions.
- `C1` The revision pack covers both stated lesson objectives completely and matches the curriculum target for JSS1 3D shapes at the expected depth. No objective is omitted or superficially treated.
- `C2` The pack covers both stated lesson objectives completely and in depth, matching the curriculum target for JSS1 Third Term week 4 on three-dimensional shapes. Nothing required by the objectives is missing.
- `C2` The revision pack directly and completely addresses both stated lesson objectives, covering every shape named in the lesson plan and matching the depth expected at JSS1 level.
- `C2` The revision pack covers both stated lesson objectives fully and in the correct order, matching the JSS1 Third Term curriculum target for 3D shapes. Nothing relevant is omitted.
- `C2` The revision pack addresses both stated lesson objectives completely and in appropriate depth for JSS1 Third Term, leaving no curriculum target unaddressed.

- `C2` The pack covers both stated lesson objectives completely and in the correct order, matching the JSS1 Third Term curriculum target on 3D shapes. Nothing essential is missing, and no out-of-scope objectives are introduced.
- `C2` The pack covers both stated lesson objectives completely and directly, matching the curriculum target for JSS1 Term 3 Week 4 on 3D shapes with no gaps.
- `C3` The pack covers both stated lesson objectives completely and directly mirrors the teacher's evaluation questions, giving full alignment with the curriculum target for JSS1 Term 3 Week 4.
- `C3` Both stated lesson objectives are fully addressed in the revision pack, and the content matches the JSS1 Third Term curriculum expectation for 3D shape properties. Nothing of the curriculum target is omitted.
- `C3` The pack covers both stated objectives completely and also mirrors the teacher's own evaluation questions, showing tight alignment with the intended curriculum target for JSS1 Term 3 Week 4.
- `C3` Both stated lesson objectives are addressed comprehensively and in the correct JSS1 scope. All teacher evaluation questions find a counterpart in the revision pack's practice or explanatory text.
- `C3` The revision pack fully addresses both stated lesson objectives and the curriculum content for JSS1 3D shapes, covering all required solids and their properties with nothing important omitted.
- `C3` Both stated lesson objectives are addressed completely and in order. The pack also includes the exact types of questions the teacher set for evaluation, confirming strong alignment with the curriculum target for JSS1 Term 3 Week 4.

Taught-lesson alignment — C0 3.67, C1 4.00, C2 4.00, C3 4.00

- `C0` Sequence and emphasis match the taught lesson closely. The minor gap is the absence of any treatment of a hexagonal-base pyramid (which the teacher assessed) and the tetrahedron (5 vertices question), so alignment is strong but not complete.
- `C0` While the topic sequence is preserved, the pack adds formula-based reasoning and worked examples that exceed the lesson's described depth, and it omits the drawing/labelling activities that featured in the teacher's own evaluation.
- `C0` The pack mirrors the lesson sequence and covers the same content at roughly the right depth, but the absence of any drawing/labelling activity (which the teacher explicitly evaluated) and the richer worked-example format mean alignment is strong but not perfect.
- `C0` The pack mirrors the lesson's sequence and conceptual emphasis well, and addresses the same properties the teacher tested. It loses one point because the drawing/labelling activities that formed part of the teacher's evaluation are not present in the pack.
- `C0` While the pack respects the lesson's topical sequence and depth, it omits the drawing activities that were central to the lesson's method and evaluation. This is a meaningful gap between what was taught and what the revision pack requires students to do.
- `C0` The pack mirrors the lesson's sequence and depth faithfully and echoes most of the teacher's evaluation questions, though the drawing-and-labelling activity from the lesson evaluation is absent, slightly reducing alignment.
- `C1` The pack matches the lesson's sequence and covers all the content the teacher presented, and the practice questions align closely with the teacher's own evaluation. The shift from board-copy to structured explanation is appropriate for home use and does not represent a significant misalignment.
- `C1` The pack mirrors the lesson's sequence and directly reuses the teacher's evaluation questions, showing strong alignment. The one gap is the absence of a drawing/labelling task, which the teacher included as evaluation question 3.
- `C1` The pack reflects the lesson's sequence and covers the exact content the teacher assessed. The formula-based approach extends slightly beyond the simple note-taking method described, but this is a helpful rather than harmful addition, and the core depth and emphasis match well.
- `C1` The structural sequence and emphasis closely mirror the lesson plan, and the practice activities echo the teacher's evaluation questions; the slightly richer explanatory depth is a minor deviation from the taught method but does not misrepresent the lesson content.
- `C1` The pack reflects the lesson's sequence and counting emphasis well, but the drawing/labelling activities that featured in both the lesson steps and the teacher's own evaluation are absent, which is a moderate gap in taught-lesson alignment.
- `C1` The pack's structure and emphasis closely match the taught lesson's order and focus, and it incorporates all five evaluation questions. The algebraic generalisation is a small addition beyond the board-and-notes method described, which prevents a perfect score but does not seriously misrepresent the lesson.
- `C2` The pack faithfully mirrors the lesson sequence and emphasis, and closely reproduces the teacher's evaluation questions as practice items. The one minor departure is the explicit formula for pyramid properties, which enriches but slightly exceeds the stated taught depth.

- `C2` The pack closely follows the lesson's sequence and content emphasis, and its practice questions align well with the teacher's evaluation questions, but the absence of a drawing-and-labelling exercise means one explicit classroom activity is not carried over.
- `C2` The pack closely follows the lesson's sequence and depth, and most teacher evaluation questions are represented. The one gap is the absence of a drawing/labelling task, which the teacher explicitly included in the evaluation, so alignment is very good but not perfect.
- `C2` The pack closely follows the lesson's sequence and covers the same shapes and properties the teacher taught, but it gives less emphasis to the drawing-and-labelling activity that was central to the classroom lesson.
- `C2` The pack is strongly aligned with the taught lesson's sequence and assessment questions. The small additions (vocabulary box, pyramid table, worked example) enrich rather than distort the lesson, so only a minor deduction is warranted.
- `C2` The pack faithfully reproduces the lesson's sequence and note-taking/counting approach, and echoes most evaluation questions, though it substitutes a pentagonal pyramid for the teacher's hexagonal example and adds the n-sided formula pattern, which goes slightly beyond what the bare lesson plan suggests.
- `C3` The pack faithfully reproduces the lesson's sequence and depth, and practice questions align closely with the teacher's evaluation questions; the slight gap is that the hexagonal pyramid appears only in practice rather than in the expository content, but this is a small omission.
- `C3` The pack faithfully reproduces the lesson's sequence and depth; the only gap is that the teacher asked students to draw and label shapes, while the pack provides figure descriptions rather than asking students to draw — a minor divergence in activity type.
- `C3` The pack closely follows the lesson's sequence and note-taking methodology; the only minor gap is that the hexagonal base pyramid is treated via the algebraic pattern rather than as a drawing exercise as the teacher's evaluation required, but this is a small deviation.
- `C3` The pack mirrors the lesson's order, content and expected depth closely. A minor gap is that the lesson explicitly mentioned cylinders and spheres among instructional materials but the pack does not address them — though those shapes were not in the objectives, so this is a minor misalignment rather than a serious one.
- `C3` The pack mirrors the lesson's sequence and depth well and reflects the teacher's assessment approach, but it does not replicate the drawing-and-labelling activity that was part of both instruction and evaluation, so alignment is strong but not perfect.
- `C3` The sequence and emphasis match the taught lesson closely, and the content depth is appropriate for JSS1. The formula and table are a modest extension beyond what the lesson plan records, but they support rather than contradict the lesson, so alignment is strong though not perfect.

Scope and conceptual correctness — C0 4.67, C1 5.00, C2 4.83, C3 4.83

- `C0` No conceptual or factual errors were found anywhere in the pack. All counts and descriptions stay within JSS1 scope and are mathematically sound.
- `C0` The mathematical content is accurate throughout, with correct counts for all shapes and a valid generalisation for pyramids; the only minor issue is slight inconsistency in terminology around the cone's curved surface, which does not constitute a conceptual error.
- `C0` No mathematical or conceptual errors were found anywhere in the pack; all counts and formulas are accurate, and the pack stays within the lesson boundary without introducing out-of-scope material.
- `C0` Every property stated in the pack is mathematically correct and within the expected scope of JSS1 work on 3-D shapes. No errors, over-complications, or out-of-scope concepts were found.
- `C0` There are no conceptual or numerical errors in the pack. The one slight scope concern is the introduction of general algebraic formulae for pyramids (faces = 1+n, etc.), which goes slightly beyond a typical board-and-notes JSS1 lesson, but this is a minor and helpful extension rather than an error.
- `C0` The pack contains no mathematical or conceptual errors across all four shapes. The scope stays within the JSS1 lesson boundary without straying into unrelated solid geometry topics.
- `C1` Every mathematical claim in the pack is accurate and the answer key is error-free. The pack stays within the scope of the lesson without introducing content beyond JSS1 expectations.
- `C1` All numerical facts and conceptual statements are mathematically correct. The scope stays within JSS1 boundaries and does not introduce material beyond the lesson.
- `C1` No mathematical or conceptual errors were found anywhere in the pack. Every property and worked example has been verified as correct, and the scope stays within 3D shapes appropriate for JSS1.

- `C1` Every mathematical claim in the pack is accurate, the answers in the 'Check your work' section are all correct, and the content remains appropriately within the JSS1 three-dimensional shapes boundary.
- `C1` Every numerical property and conceptual claim in the pack is mathematically correct and confined to the scope appropriate for a JSS1 lesson on 3-D shapes; no errors or out-of-scope content were found.
- `C1` Every mathematical fact and count in the pack is correct, and the scope stays within JSS1 3D-shapes content without straying into unrelated topics or introducing errors.
- `C2` There are no conceptual or mathematical errors anywhere in the pack. All definitions, counts, and formulae are accurate, and the scope stays within the boundaries of JSS1 three-dimensional shapes.
- `C2` Every mathematical fact in the pack is accurate and consistent with the JSS1 curriculum scope; no errors or out-of-scope content were found.
- `C2` Every mathematical fact, formula, and count in the pack is accurate and stays firmly within the JSS1 lesson boundary. No conceptual errors or out-of-scope content were found.
- `C2` No mathematical or conceptual errors were found anywhere in the pack; scope stays within the JSS1 lesson boundary without straying into unrelated or advanced material.
- `C2` Every numerical property and conceptual statement in the pack is mathematically correct. The pack stays within the expected JSS1 lesson boundary and introduces no errors or misconceptions.
- `C2` The content is conceptually accurate throughout; the small ambiguity in answer 4 about square vs rectangle is technically correct and does not constitute an error, keeping the score high but short of perfect.
- `C3` No mathematical or conceptual errors were found anywhere in the pack; all counts and definitions are correct and appropriately scoped for JSS1.
- `C3` Every numerical claim in the pack is mathematically correct and no concept is taken beyond the expected JSS1 boundary; the general rule linking a pyramid's base polygon to its face/edge/vertex counts is a useful but appropriately scoped extension.
- `C3` The mathematical content is essentially correct throughout; the only concern is the informal and slightly imprecise use of 'regular/irregular' for solid shapes in the introduction, which is beyond standard JSS1 terminology and could mildly confuse students.
- `C3` No mathematical or conceptual errors were found anywhere in the pack. The content stays within the JSS1 three-dimensional shapes boundary without straying into surface area, volume, or other topics beyond the lesson scope.
- `C3` There are no mathematical or conceptual errors in the pack; all shape properties are accurately stated and the scope stays within what is expected at JSS1 level without straying into topics beyond the lesson.
- `C3` Every mathematical fact and count in the pack is accurate, the formula is correctly derived and applied, and the scope stays within the JSS1 three-dimensional shapes topic without straying into unrelated content.

Student-facing clarity and usefulness — C0 5.00, C1 5.00, C2 5.00, C3 5.00

- `C0` The pack is exceptionally well-structured for independent home use: clear definitions, relatable analogies, guided examples, and fully explained answers make it highly usable by a JSS1 student working alone.
- `C0` The pack is exceptionally well structured for independent home use: language is simple and relatable, worked examples model the expected reasoning, and the answer key allows self-checking — exactly what a take-home revision pack needs.
- `C0` The pack is exceptionally well-suited for independent home use: plain language, relatable objects, clearly structured examples, and fully explained answers give a JSS1 student everything needed to review and self-assess without adult help.
- `C0` The pack is written in accessible, age-appropriate language with concrete illustrations, structured examples, and self-check answers — exactly what a student needs for unsupported home revision.
- `C0` The pack is exceptionally well-structured for independent home use: language is plain, examples are relatable, worked solutions model the thinking step by step, and full answers with explanations are provided so students can self-check.
- `C0` The language is accessible and age-appropriate, the structure moves logically from explanation to example to practice to answers, and the pack is self-contained enough for a JSS1 student to use independently at home.
- `C1` The pack is exceptionally well structured for independent home use: vocabulary is introduced before it is needed, worked examples precede practice questions, and detailed answer keys support self-correction. A JSS1 learner with no teacher support should be able to follow it effectively.
- `C1` The language is plain and age-appropriate, the structure moves logically from explanation to practice to answers, and the worked solutions are genuinely instructive. A JSS1 student working alone at home can follow and use this material without difficulty.

- **C1** The language is accessible, the structure moves clearly from explanation to example to practice to self-check, and the real-world analogies and formula breakdowns make this highly usable for an independent JSS1 learner revising at home.
- **C1** The pack is clearly structured for independent home use: vocabulary is introduced accessibly, worked examples model the expected reasoning, practice questions match the lesson's assessment level, and complete answer explanations allow self-correction — all highly appropriate for a JSS1 learner.
- **C1** The language is clear and age-appropriate, concrete examples ground each concept, and the self-contained answer key makes this pack highly usable as standalone home-revision material for JSS1 students.
- **C1** The language is plain and age-appropriate, the structure moves logically from explanation to examples to practice to answers, and every section a JSS1 student needs to work alone at home is present and usable.
- **C2** The pack is exceptionally well-structured for independent home use: clear definitions, worked models, graded practice questions, and self-check answers all support a JSS1 learner without teacher assistance. Language is simple and appropriate for the age group.
- **C2** The pack is well-structured, uses accessible language appropriate for JSS1, and provides everything a student needs to review, practise, and self-check, making it genuinely useful as standalone take-home material.
- **C2** The pack is exceptionally well-structured for independent home use: definitions, figures, worked examples, practice questions and answers all appear in logical order and use language appropriate for JSS1 learners.
- **C2** The pack is well-structured, uses accessible language appropriate for JSS1, and provides all the scaffolding a student needs to review and self-check independently without a teacher present.
- **C2** The pack is clearly written, logically structured, and contains all the scaffolding a JSS1 learner needs to review the lesson independently. The combination of worked examples, practice questions, and an answer key with explanations makes it genuinely useful as take-home material.
- **C2** The pack is exceptionally well structured for independent home use: definitions precede tasks, worked examples scaffold practice, and fully explained answers enable self-assessment — all at language and conceptual levels appropriate for JSS1 learners.
- **C3** The pack is well-structured, uses age-appropriate language, and supplies enough scaffolding (definitions, worked reasoning, answer explanations) that a JSS1 student can meaningfully review the lesson at home without teacher support.
- **C3** The pack is well-structured, uses accessible language, and provides sufficient scaffolding and answer keys for a JSS1 learner to review the lesson and check understanding without a teacher present.
- **C3** The pack is well-structured for independent home use: definitions come before exercises, figures are described, a reference table is included, and every answer is explained — all features that support a JSS1 student working alone.
- **C3** The pack is well-structured for independent home use: definitions precede exercises, figures are described in enough detail to be useful, and fully explained answers allow a student to diagnose their own errors without a teacher present.
- **C3** The pack is clearly written, logically structured, and self-contained with answers and explanations, making it highly suitable for JSS1 learners to use independently as a take-home revision resource.
- **C3** The pack is well-structured for unsupported home use: definitions, worked examples, practice and answers all appear in a logical order with accessible language throughout, making it genuinely useful as take-home revision material for JSS1 students.